CHEM 1301 Driller 9-2

Section 8.7 (empirical and molecular formulas) + Chapter 9 — 15 exam-style multiple-choice questions

Instructions. Choose the single best answer (a–e) for each question. You may use a periodic table. Calculators are allowed. Answers are rounded to the precision shown. Time yourself: aim for about 30 minutes (roughly 2 minutes per question, the same pace as the full exam).

Answer sheet. Use the bubble answer sheet (`CHEM1301/exam-prep/answer-sheet/answer-sheet.pdf`), rows 1–15.

1. Which statement about empirical and molecular formulas is correct?

a) The molecular formula is always a whole-number multiple of the empirical formula
b) The empirical formula gives the actual number of atoms in one molecule
c) The empirical formula is always different from the molecular formula
d) An empirical formula may contain fractional subscripts such as $CH_{1.5}$
e) Two different compounds can never have the same empirical formula

2. Hexane has the molecular formula C_6H_{14}. What is its empirical formula?

a) CH_2
b) C_2H_5
c) C_6H_{14}
d) C_3H_7
e) C_3H_8

3. A nitrogen oxide contains 25.94% N and 74.06% O by mass. What is its empirical formula?

a) NO_5
b) N_2O_5
c) NO_3
d) N_4O_{10}
e) NO_2

4. Ethylene glycol is 38.70% C, 9.74% H, and 51.56% O by mass. What is its empirical formula?

a) CH_3O
b) CH_4O
c) C_4HO_5
d) C_2H_3O
e) $C_2H_6O_2$

5. A 5.00 g sample of a sulfur oxide contains 2.00 g of sulfur; the rest is oxygen. What is its empirical formula?

a) S_2O_6
b) SO_2
c) S_3O
d) SO_3
e) S_2O_3

6. Butane has the empirical formula C_2H_5 and a molar mass of 58.12 g/mol. What is its molecular formula?

a) C_4H_5
b) C_8H_{20}
c) C_4H_{10}
d) C_6H_{15}
e) C_2H_5

7. A gaseous hydrocarbon is 85.63% C and 14.37% H by mass, and its molar mass is 28.05 g/mol. It burns completely in O_2 to form CO_2 and H_2O. How many moles of O_2 are needed to burn 0.500 mol of the hydrocarbon?

a) 0.167 mol
b) 0.500 mol
c) 0.750 mol
d) 1.00 mol
e) 1.50 mol

8. The **limiting reactant** in a chemical reaction is the reactant that:

a) is left over when the reaction stops
b) is completely used up first and so determines how much product can form
c) has the fewest moles present
d) has the smallest coefficient in the balanced equation
e) has the smallest mass

9. Air bags inflate by the reaction $2NaN_3(s) \rightarrow 2Na(s) + 3N_2(g)$. How many moles of N_2 form from 1.50 mol of NaN_3?

a) 1.00 mol
b) 1.50 mol
c) 2.25 mol
d) 9.00 mol
e) 63.0 mol

10. For $N_2(g) + O_2(g) \rightarrow 2NO(g)$, what mass of NO forms from 0.750 mol of N_2 with excess O_2?

a) 1.50 g
b) 11.3 g
c) 22.5 g
d) 42.0 g
e) 45.0 g

11. Zinc dissolves in acid: $Zn(s) + 2HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$. What mass of H_2 forms when 10.0 g of Zn reacts with excess HCl?

a) 0.153 g
b) 0.308 g
c) 0.617 g
d) 10.0 g
e) 13.2 g

12. Silicon reacts with chlorine according to the **unbalanced** equation $Si(s) + Cl_2(g) \rightarrow SiCl_4(l)$. What mass of Cl_2 is needed to react completely with 7.00 g of Si?

a) 0.498 g
b) 8.83 g
c) 14.0 g
d) 17.7 g
e) 35.3 g

13. Methane burns: $CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g)$. A mixture of 8.00 g of CH_4 and 24.0 g of O_2 reacts. Which statement is correct?

a) O_2 is limiting, and 1.98 g of CH_4 is left over
b) O_2 is limiting, and 6.02 g of CH_4 is left over
c) Neither is limiting; both reactants are used up completely
d) O_2 is limiting, and 8.00 g of CH_4 is left over
e) CH_4 is limiting, and O_2 is left over

14. For $2NO(g) + O_2(g) \rightarrow 2NO_2(g)$, 15.0 g of NO reacts with 10.0 g of O_2. What is the theoretical yield of NO_2?

a) 0.500 g
b) 15.0 g
c) 23.0 g
d) 25.0 g
e) 28.8 g

15. Baking soda decomposes when heated: $2NaHCO_3(s) \rightarrow Na_2CO_3(s) + H_2O(g) + CO_2(g)$. Heating 16.8 g of $NaHCO_3$ gives 9.85 g of Na_2CO_3. What is the percent yield?

a) 0.929%
b) 46.5%
c) 58.6%
d) 92.9%
e) 108%